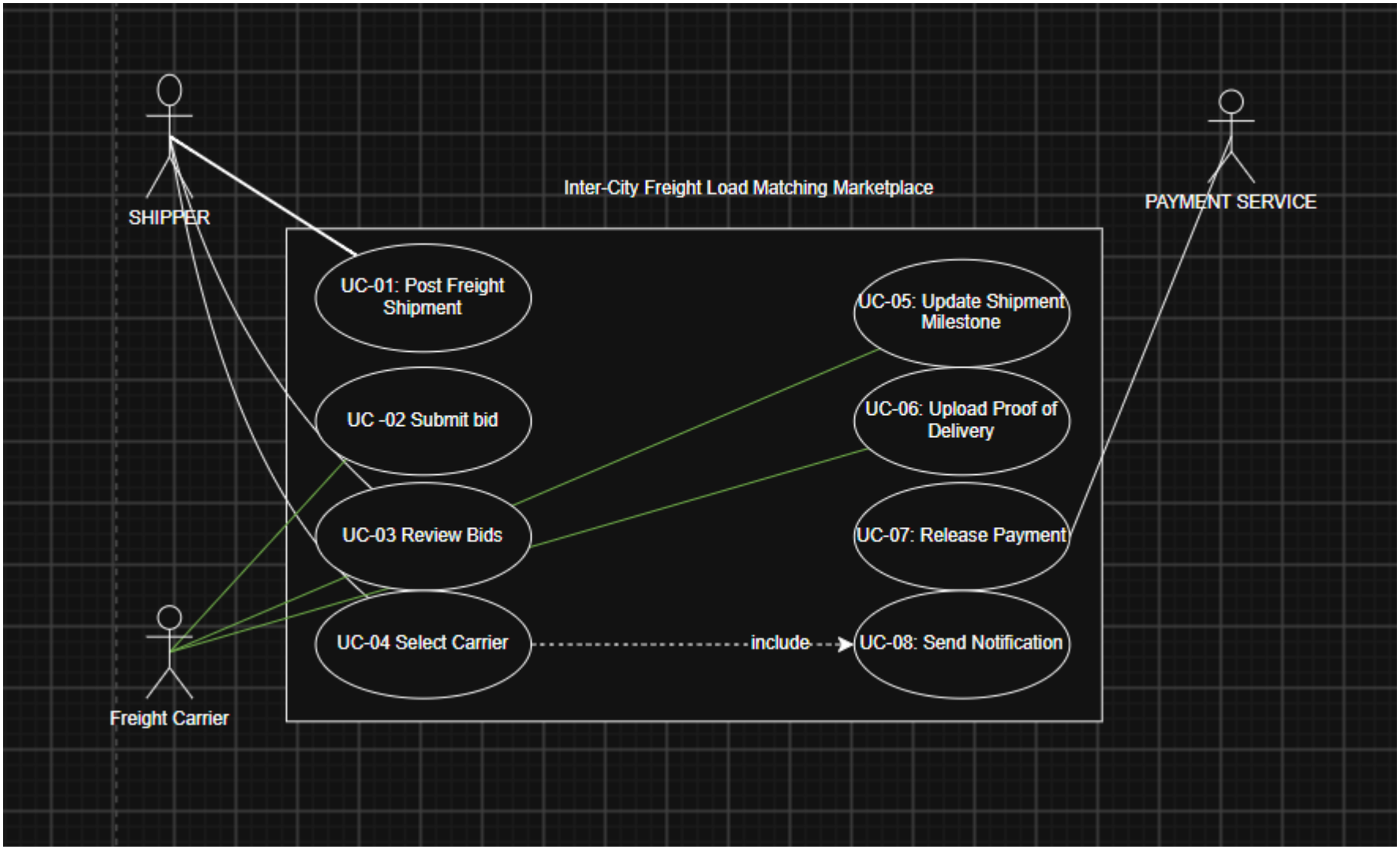
Problem Statement #28 — Inter-City Freight Load Matching Marketplace

Lab 1 — Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.	High	Pass: Carrier bid recorded and shipper notified. Fail: Unverified carrier submits bid without valid permits.	Core marketplace function; ensures only credentialed carriers compete for loads.	Given
FR-002	Functional	The system shall allow shippers to post freight load details, including origin, destination, cargo type, weight, and pickup window.	High	Pass: Load appears in the active marketplace listing with all mandatory fields populated. Fail: Load with a missing mandatory field is rejected with a validation error.	Enables carriers to discover and evaluate available freight before bidding.	
FR-003	Functional	The system shall allow shippers to review submitted bids and award the load to a selected carrier before the auction closes.	High	Pass: Selected bid status changes to "Awarded" and all other bidders are notified of non-selection. Fail: Load remains open after shipper confirms a selection.	Finalizes the shipper-carrier match and is the trigger for shipment execution.	
FR-004	Functional	The system shall allow carriers to submit proof-of-delivery (e-signature, photo, GPS geotag, and timestamp) upon completion of a delivery.	High	Pass: A POD submission with all required evidence is timestamped and locked from further edits. Fail: Submission missing e-signature or geotag is rejected.	Proof-of-delivery is the trigger event for milestone payment release and dispute resolution.	
FR-005	Functional	The system shall notify shippers and carriers in real time of load status changes (posted, bid awarded, in-transit, delivered, payment released).	Medium	Pass: Push/email notification is delivered to affected users within 5 seconds of the status change. Fail: A status change occurs without any notification being sent.	Keeps both parties informed and reduces support inquiries about shipment status.	
NFR-001	Nonfunctional	Milestone payments held in escrow shall disburse automatically within 60 seconds of verified electronic proof-of-delivery submission.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Fast, reliable payment release builds trust between shippers and carriers.	Given (Type: Performance & Security)
NFR-002	Nonfunctional	The system shall encrypt all carrier verification (KYC) data and payment transaction data using TLS 1.2+ in transit and AES-256 at rest.	High	Pass: Security audit confirms all data-in-transit and data-at-rest channels use approved encryption standards. Fail: Any unencrypted transmission or storage of sensitive data is detected.	Protects sensitive financial and business data and supports regulatory compliance.	Type: Security

USE CASE DIAGRAM



– Use-Case Flow Specification

Use Case ID: UC-02
Use Case Name: Submit Bid
Primary Actor: Freight Carrier
Supporting Actor: Shipper
Preconditions

1. A shipper has already posted a freight shipment.
2. The shipment is available for bidding.
3. The freight carrier is verified and authorized to submit bids.

Postconditions

1. The valid carrier bid is recorded against the freight shipment.
2. The shipper is notified that a new bid has been received.

Main Success Scenario

1. The freight carrier views the available freight shipments.
2. The carrier selects an open freight shipment.
3. The system displays the shipment details.
4. The carrier enters the proposed bid amount and required bid details.
5. The system validates the carrier and the submitted bid information.
6. The carrier submits the bid.
7. The system records the bid for the selected shipment.
8. The system confirms that the bid has been successfully submitted.
9. The system notifies the shipper about the new bid.
10. The use case ends successfully.

Alternate Flow – Invalid Bid Information

5a. The system detects missing or invalid bid information.

1. The system displays an error message indicating the information that needs correction.
2. The carrier corrects the invalid or missing information.
3. The carrier resubmits the bid.
4. The system validates the corrected bid.
5. If the bid is valid, the system records it and notifies the shipper.

The flow continues from Step 7 of the Main Success Scenario